CSU | Washkewicz College of Engineering

Quantum Computing: Hands-On Introduction

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Are Quantum Computers "Faster"?

- p*q = N (multiplication)
  - p and q are 2048-bit integers
  - Classical computer: 0.0025 seconds
  - Quantum computer: 75 seconds
- N = p*q (factorization)
  - p and q are 2048-bit integers
  - Classical computer: 4.7 billion CPU years
  - Quantum computer: 8 hours

Quantum computers can solve certain problems very, very fast.
But they won't replace classical computers.

Qubits

- A qubit can take the value of 0, 1, or both at the same time with probability.
- A qubit is in a “superposition” state.

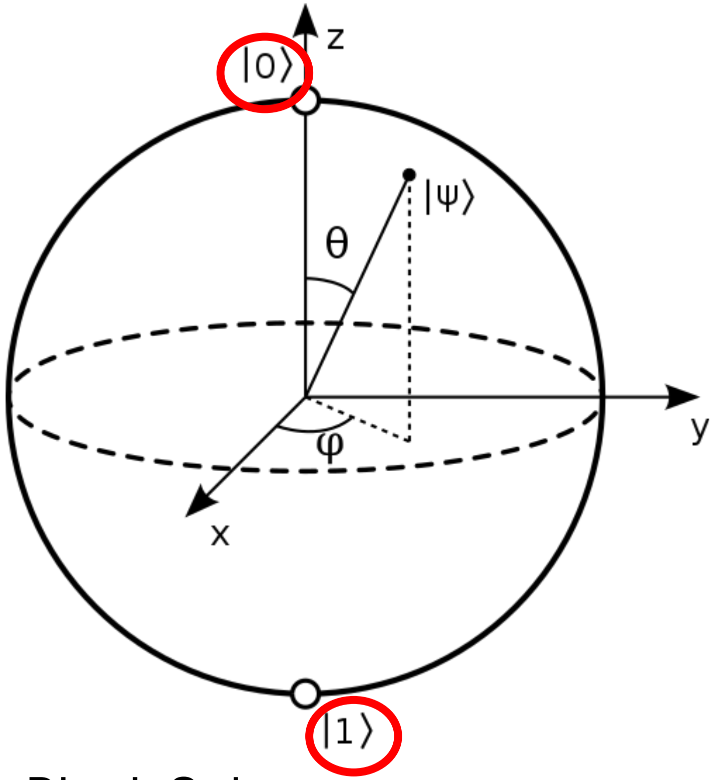
If measured, this qubit will collapse to
0 with probability $|a|^2$ and
1 with probability $|b|^2$, where $|a|^2+|b|^2 = 1$.

This qubit is denoted as:

$\begin{pmatrix} a \\ b \end{pmatrix}$ **or** $a|0> + b|1>$

Qubit: Visualization

$\binom{a}{b}$ = a ·|0> + b ·|1>, where $|a|^2+|b|^2 = 1$



Bloch Sphere

https://en.wikipedia.org/wiki/Bloch_sphere

A qubit is represented as a point on the surface of the Bloch Sphere.

$$|\psi\rangle = \underbrace{\cos\left(\frac{\theta}{2}\right)}_{a}|0\rangle + \underbrace{e^{i\varphi}\sin\left(\frac{\theta}{2}\right)}_{b}|1\rangle$$

$(|a|^2+|b|^2 = 1)$

Qubit: Visualization

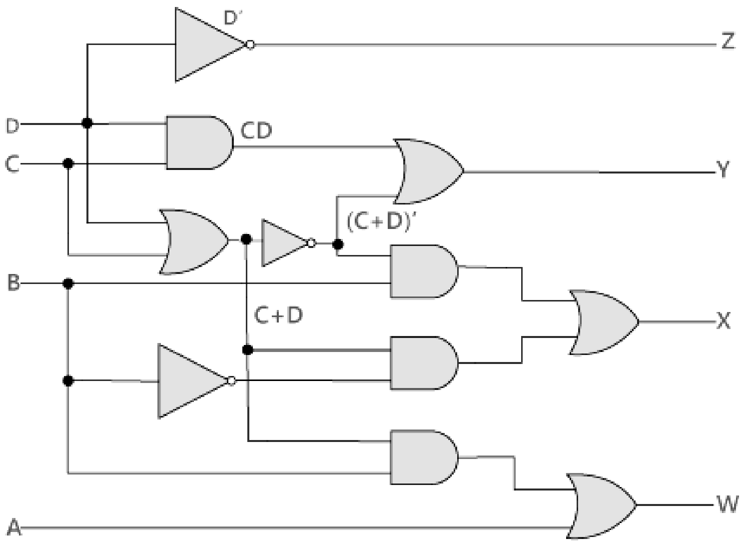
- Bloch Sphere Simulator (https://bloch.kherb.io/)
  - |0> and |1>
  - X gate (operator or transformation)
  - H gate
  - 90, 45, 60 & Z gate

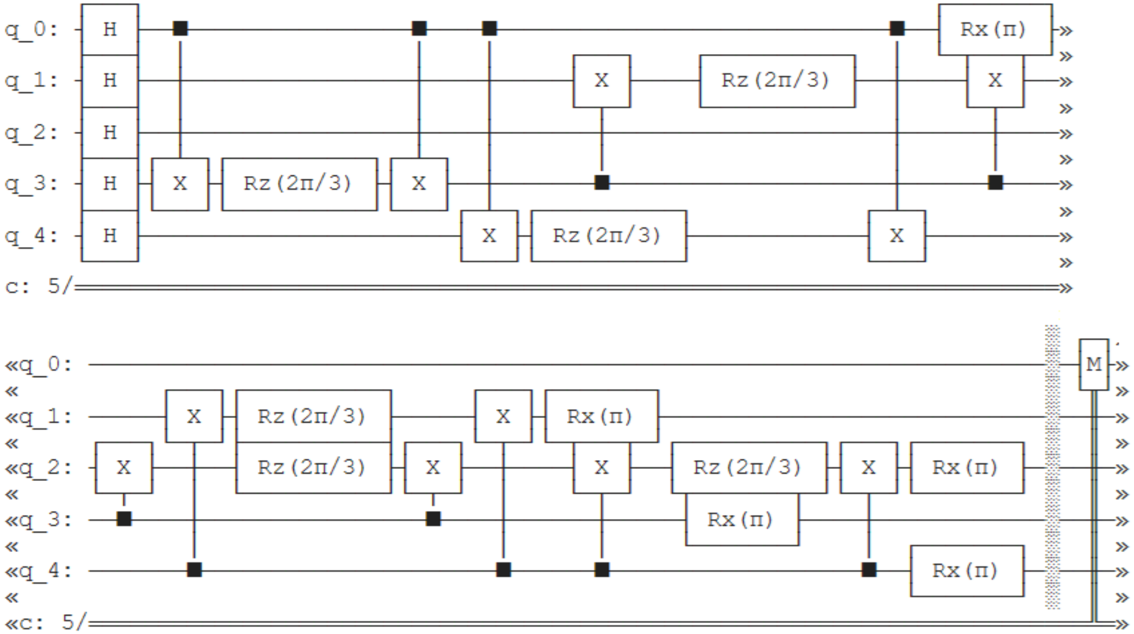
  - Sometimes, a problem can easily be encoded in "phase"

Exercise Question: How does S and T gate transform/rotate a qubit?

Classical vs Quantum Gates and Circuits

- Classical logic gates are basic building blocks for digital circuits.
- Quantum gates are building blocks for quantum circuit, through which qubits evolve to compute what we want.





Quantum Circuit Simulator

- Guessing a 4-bit PIN (not “digit”, for simplicity)
  - Total PINs: 16
  - Classical guessing: 8 tries on average
  - Quantum guessing: a couple of tries

    (try 16 PINs all at once)

- QC Simulator (https://algassert.com/quirk)
  - 4H
  - 4H, (2X, CCCZ, 2X), assuming PIN is “0110”
  - 4H, (2X, CCCZ, 2X), (4H, 4X, CCCZ, 4X, 4H)
  - Same as above: 4H, Marking, Amplification

Exercise Question: What if another “Marking” and “Amp.” are added?

IBM Quantum System

- Compute resources in the cloud (IBM) (https://quantum.cloud.ibm.com/computers)
    - ibm_kingston, ibm_pittsburgh, ibm_torino, ibm_cleveland, …
    - Obtain CRN and API key
- Run (https://colab.research.google.com/)
    - Google Colab (Python and Jupyter Notebook)

https://www.ibm.com/quantum/blog/open-plan-updates

Challenges: Noise, Scalability …

- How long a qubit can hold its quantum information?
- Quantum gate error; Readout error

- E.g., ibm_fez
  - T_1: Excited state |1> to ground state |0> by losing energy to its environment, 132.07 usec
  - T_2: Losing its phase information, 97.22 usec
  - 2Q error: two-qubit gate error, 0.263%
  - Readout error: 0.879%

Questions?

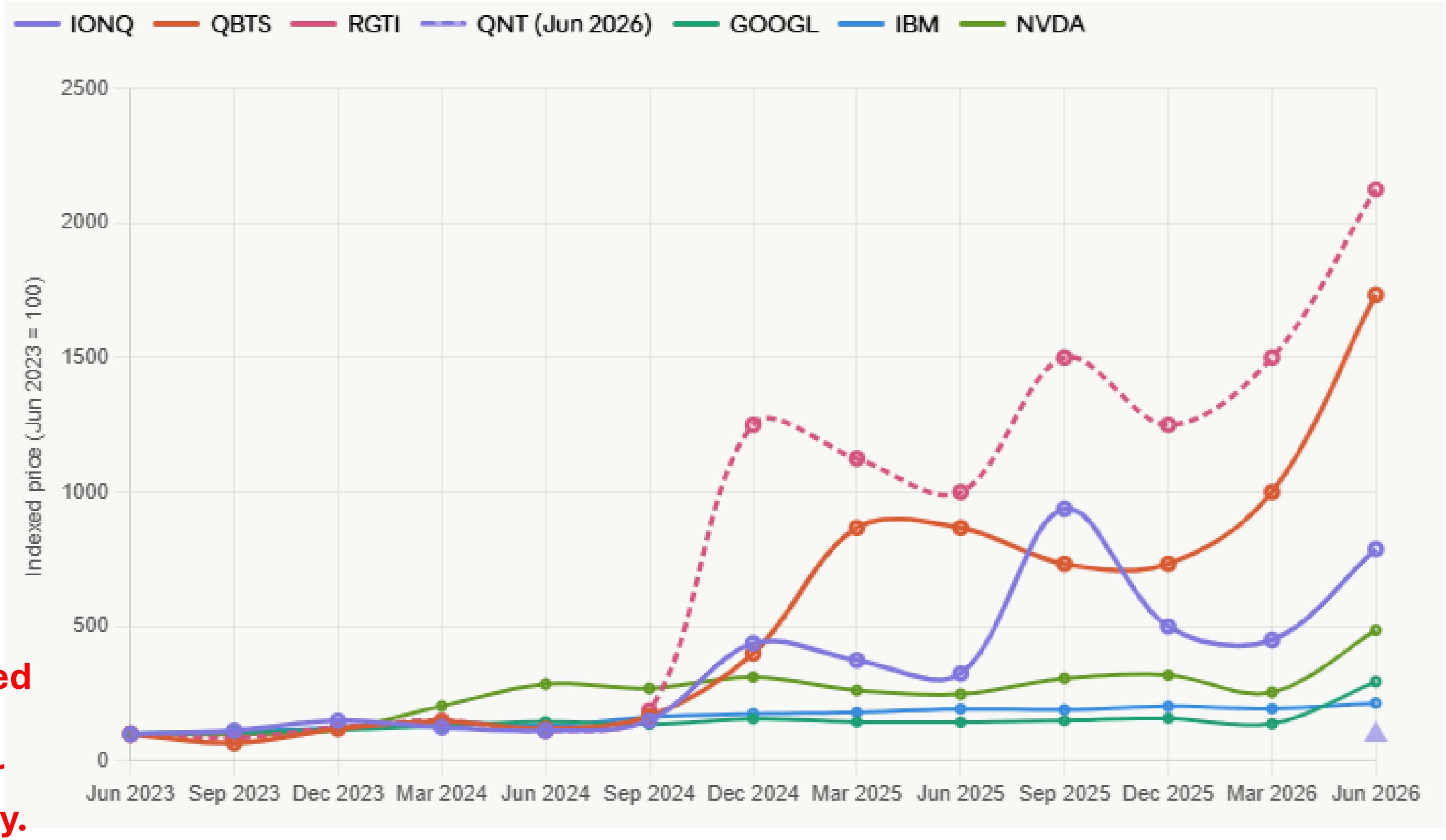
When will it be available?
One Wright brother said to the other that **man will not fly for a thousand years**; **two years later they actually flew** the Wright Flyer. Fermi said that net energy from nuclear reactions is a 50-year matter; **two years later he personally oversaw building the first nuclear pile**. (Eliezer Yudkowsky, Hard Fork episode on 9/12/25)

Very confusing, isn't it? "If you think you understand quantum mechanics, you don't understand q.m."
"Behind it all is surely an idea so simple, so beautiful, that when we grasp it - in a decade, a century, or a millennium - we will all say to each other, **how could it have been otherwise? How could we have been so stupid?**" (John A. Wheeler)

It's a hype which will come and go, right?
"**If we ever meet aliens, my bet is that they'll use quantum computers** (or, perhaps, will have quantum computing brains)." What's more, Deutsch pointed out, you wouldn't need to rely on informal, heuristic arguments to justify your notion of algorithm, as Turing had done. You could use the laws of physics to prove your device was universal. (by Andy Matuschak and Michael Nielsen)

* For any questions, please contact Dr. C. Yu at c.yu91@csuohio.edu
* If you're interested, please enroll in the micro-credential course on Quantum Computing, https://www.csuohio.edu/continuing-education/intro-quantum-computing

Three-Year Trend of Quantum Companies



Prices indexed to 100 at Jun 2023 start for comparability.

Quantinuum (QNT) IPO'd Jun 4, 2026 at $60.

Quantum Computing at Cleveland State

- Courses
  - EEC 489 / 589 Quantum Computing (QC)
  - EEC 459 / 559 Quantum Information Science (QIS)
  - CIS 492 Quantum Machine Learning (QML)
- Micro-credential
  - Introduction to Quantum Computing (2nd cohort ongoing; 3rd cohort is being planned)
  - Quantum Optimization and Simulation (starts on October 12, 2026)
  - Quantum Key Distribution and Error Correction/Mitigation (2027)
- Bachelor of Science in Computer Engineering, Quantum Computing track
- PhD in Applied Data Science with Quantum Computing
  - A new joint PhD program with Cleveland Clinic, where Quantum Computing is one of the focus areas